

Introduction

The Kotlin language has been around since 2011, but its popularity took off in 2017 when Google announced Kotlin's inclusion as a first-class language for Android development. In 2019, Google announced a "Kotlin-first" approach to Android development. With modern and expressive language characteristics such as those found in Apple's Swift, and 100% interoperability with Java, it's no wonder that Kotlin has been named a top 5 most-loved language by Stack Overflow users.

If you're a complete beginner to programming, this is the book for you! There are short exercises and challenges throughout the book to give you some programming practice and test your knowledge along the way.

Through *Kotlin Apprentice*, you'll learn about basic things like constants, values and types, move up to more complicated items such as data structures, classes and enumerators, and finish off with some in-depth knowledge about functional programming, coroutines, and Kotlin Multiplatform.

If you want to get right into Android app development after you work through *Kotlin Apprentice*, we suggest you read the *Android Apprentice*, available on our store:

- <https://www.raywenderlich.com/books/android-apprentice>

The *Android Apprentice* is your introduction to building great apps in Android, using the Kotlin language, for both novice programmers and those with extensive experience programming for iOS or other platforms.

It will help you master the essential building blocks of Kotlin and Android to start creating apps. As you work on more apps, you'll find the foundations you learn in *Kotlin Apprentice* and *Android Apprentice* will give you the knowledge you need to easily figure out more complicated details on your own.

Section I: Kotlin Basics

The chapters in this section will introduce you to the very basics of programming in Kotlin. From the fundamentals of how computers work, all the way up to language structures, you'll cover enough of the language to be able to work with data and organize your code's behavior.

The section begins with some groundwork to get you started, then you have a chance to *do* things with your newly learned data types. Finally, you'll move past the variables and constants you've been working with, which have had concrete values, to more tricky data types.

These fundamentals will get you on your way and, before you know it, you'll be ready for the more advanced topics that follow. Let's get started!